

It is Easy to Make Your Products IoT-Ready

Thingsquare Mist is a game-changing software system that brings true Internet connectivity to the Internet of Things

Thingsquare, a company founded in 2012 by Contiki's creator Adam Dunkels, works towards simplifying the Internet of Things (IoT), with open source platforms that help add wireless mesh networking and Internet connectivity to any and every product. Their latest software platform, Thingsquare Mist, can be used in connected-home products, smart lighting systems and smart city projects.

It is so easy to use that any home owner can use it to Internet-enable home equipment and appliances. Also, one can use existing Internet equipment without the hassle of USB dongles or costly home router replacements. In large buildings such as offices and hotels, Thingsquare Mist mesh technology enables large networks without having to extend the Wi-Fi network. With Thingsquare Mist, it is also possible to connect smart lighting systems to smartphone apps for energy saving. It is also possible to Internet-connect entire cities using battery-powered wireless sensors, which can be used for smart parking, lighting, security or other purposes.

Mist builds on well-known open source components, has a small memory footprint and low computational requirements, and works with multiple microcontrollers with a range of radios. At the heart of this product is the 'sleepy mesh' technology, which

helps increase reliability and range in wireless sensor networks without sacrificing battery life.

Thingsquare Mist is supported on a range of hardware platforms, so manufacturers have multiple options including system-on-a-chip (SoC) solutions and standalone microcontrollers with radio transceivers. Both 2.4GHz radios and sub-GHz radios are supported. For example, Mist supports STMicroelectronics' SPIRITI, which is a very low-power RF transceiver, intended for RF wireless-sensor node applications in the sub-1GHz band, such as automatic metering, alarm and security systems, home and building automation, and industrial monitoring and control.

The package works on a variant of the popular Contiki operating system, and comes with an Eclipse-based development environment. All the necessary compilers and tools are combined into a single download. A simulation environment is also included so that all code can be thoroughly tested before production and deployment.

Engineers must take a look at Mist's technical merits such as the use of the world's smallest IPv6 stack, extremely small memory footprint, super-efficient mesh networking technology, frequency hopping and over-the-air firmware upgrades.

Smart soda fountain



Speaking of smart devices, think of a coke dispenser with an Intel Core i7 processor inside! Last year, SIA Interactive, a South American technology and design firm, demonstrated a coke machine with social networking features. The machine includes a webcam that can take your picture and then e-mail it to someone you choose!

Controlling what you disclose



Enterprises are now becoming increasingly connected to partner portals, mobiles, the cloud and developer communities. They need to keep a close watch and tight control over what is disclosed to whom. Understandably, application programming interface (API) management tools are now in demand as we can see from the latest spree of acquisitions—Intel acquired Mashery, CA acquired Layer 7 and so on. API management solutions like Layer 7 provide organisations the necessary tech building blocks to securely integrate applications across the new hybrid enterprise and to control how internal information is exposed externally.

A common platform for multiple devices

Zonoff, a startup that just raised \$3.8 million in Series A funding, is developing a Connected Home Platform that can link various smart devices, appliances, applications, etc to make them 'collectively smart.' Zonoff will not make its own hardware or devices. It will simply partner with consumer electronics manufacturers and home hardware manufacturers to embed technology into their devices to make them work together.

Zonoff's platform will help users to collectively control all their devices, save energy and monitor safety. It will enable manufacturers to monitor devices remotely and send technicians for service if they do not function properly. Zonoff will earn its money by licensing its technology, hosting cloud services and sharing the revenue from services offered through Zonoff.

